Counterexamples to both squared operator inequalities in MathDB 329636

Principia Math in Progress

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Task 1. Resolve the two proposed squared inequalities

For positive definite matrices, write $A\#B = A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2}$. Conjecture 2.16 of Moslehian and Fu [1] proposes that every unital positive linear map Φ, with $m_1^2 I \le A \le M_1^2 I$ and $m_2^2 I \le B \le M_2^2 I$, satisfies both

$$(\Phi(A)\#\Phi(B))^2 \le \frac{(M_1M_2 + m_1m_2)^2}{4M_1M_2m_1m_2}\,\Phi(A\#B)^2, \tag{1}$$

$$\left(\frac{M_2m_2}{M_1m_1}\Phi(A) + \Phi(B)\right)^2 \le \left(\frac{M_2}{m_1} + \frac{m_2}{M_1}\right)^2 \Phi(A\#B)^2. \tag{2}$$

Both assertions are false. We give separate exact counterexamples. All inputs are positive definite and both maps are unital and completely positive. The order is the usual positive semidefinite order.

A geometric-mean computation

Lemma 1. *Let $C > 0$, $d > 0$, $x \neq 0$, and $P_\pm = \begin{pmatrix} C & \pm x \\ \pm x^* & d \end{pmatrix} > 0$. Put $t = x^*C^{-1}x$ and $\rho = \sqrt{1 - t/d}$. Then*

$$P_+\#P_- = \operatorname{diag}(Y, d\rho), \qquad Y = C - \frac{1-\rho}{t}xx^*.$$

Proof. Congruence by $T^{-1} = \operatorname{diag}(C^{-1/2}, d^{-1/2})$ gives matrices $W_\pm = \begin{pmatrix} I & \pm w \\ \pm w^* & 1 \end{pmatrix}$, where $w = C^{-1/2}x/\sqrt{d}$. The Schur complement gives $0 < \|w\|^2 < 1$. The two matrices commute, and their product is $\operatorname{diag}(I - ww^*, 1 - w^*w)$. Thus their geometric mean is

$$\operatorname{diag}\left(I - \frac{1-\rho}{\|w\|^2}ww^*, \rho\right).$$

Congruence back by T proves the formula. Congruence invariance follows directly from the characterization of $A\#B$ as the unique positive definite X satisfying $XA^{-1}X = B$: conjugating this equation by any invertible matrix preserves the characterization. Uniqueness follows by taking the positive square root of $A^{-1/2}BA^{-1/2}$. □

Counterexample to the squared Pólya–Szegő assertion

Take $m_1 = m_2 = 1$, $M_1 = M_2 = 32$, and set

$$a = \frac{1025}{2}, \quad b = \frac{1023}{2}, \quad A = \begin{pmatrix} a & 0 & b \\ 0 & 1 & 0 \\ b & 0 & a \end{pmatrix}, \quad B = \begin{pmatrix} a & 0 & -b \\ 0 & 1 & 0 \\ -b & 0 & a \end{pmatrix}.$$

Both spectra are $\{1, 1, 1024\}$, so all spectral bounds hold. Let U interchange the first two coordinates and fix the third, and define

$$\Phi(X) = \frac{1}{4}X + \frac{3}{4}UXU^*.$$

This convex combination of unitary conjugations is unital and completely positive. Since A and B commute,

$$A\#B = \operatorname{diag}(32, 1, 32), \qquad Q := \Phi(A\#B) = \operatorname{diag}(35/4, 97/4, 32).$$

The images $\Phi(A), \Phi(B)$ are the lemma's P_+, P_- with

$$C = \frac{1}{8}\operatorname{diag}(1031, 3077), \qquad x = \frac{1023}{8}\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad d = \frac{1025}{2}.$$

Direct rational arithmetic gives

$$\rho^2 = \frac{18968594}{3251696675}, \qquad Y = C - c\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}, \qquad c = \frac{3172387}{98848}(1 - \rho).$$

Comparing squares shows $7/100 < \rho < 8/100$. Consequently

$$29 < \frac{72964901}{2471200} < c < \frac{295031991}{9884800} < 30.$$

For $z = (7, -1, 0)^T$, the lemma therefore yields

$$\|(\Phi(A)\#\Phi(B))z\|^2 = \left(\frac{7217}{8} - 4c\right)^2 + \left(\frac{3077}{8} + 12c\right)^2 > 782^2 + 732^2 = 1147348.$$

But the proposed right side of (1), evaluated on z, equals

$$\left(\frac{1025}{64}\right)^2 \|Qz\|^2 = \left(\frac{1025}{64}\right)^2 \left[\left(\frac{245}{4}\right)^2 + \left(\frac{97}{4}\right)^2\right] = \frac{36474548125}{32768} < 1147348.$$

For a Hermitian matrix H, $z^*H^2z = \|Hz\|^2$. Thus the quadratic form of the proposed right side minus the left side is negative. This disproves (1).

Counterexample to the squared Diaz–Metcalf assertion

Now use $A = \mathrm{diag}(1, 9/4, 4)$, $B = I_3$, $m_1 = 1$, $M_1 = 2$, and $m_2 = M_2 = 1$. Define

$$V = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \\ 2 & -2 \end{pmatrix}, \qquad \Psi(X) = V^* X V.$$

Since $V^*V = I_2$, this is a unital completely positive map from $\mathbb{M}_3$ to $\mathbb{M}_2$. Put $R = \Psi(A\#B)$ and $S = \frac{1}{2}\Psi(A) + I_2$. Exact multiplication gives

$$R = \begin{pmatrix} 3/2 & -1/3 \\ -1/3 & 5/3 \end{pmatrix}, \qquad S = \begin{pmatrix} 161/72 & -19/36 \\ -19/36 & 22/9 \end{pmatrix},$$

and

$$D := \frac{9}{4}R^2 - S^2 = \frac{1}{5184}\begin{pmatrix} 175 & 494 \\ 494 & 1276 \end{pmatrix}.$$

For $v = (3, -1)^T$,

$$v^* D v = -\frac{113}{5184} < 0 \qquad \left(\text{also } \det D = -\frac{1}{1296}\right).$$

This is precisely the failure of (2). If a map on one matrix algebra is preferred, replace Ψ by $\widehat{\Psi}(X) = \mathrm{diag}(V^*XV, X_{22})$ from $\mathbb{M}_3$ to itself. It is unital and completely positive, and the same negative quadratic form occurs on $(3, -1, 0)^T$.

Equality of the two bounds for B is explicitly permitted in the conjecture. The failure also persists with distinct bounds: keep $B = I_3$, and take $M_2 = 10001/10000$, $m_2 = 10000/10001$. Their product remains one, and the new quadratic form is exactly

$$v^*\left[\left(M_2 + \frac{m_2}{2}\right)^2 R^2 - S^2\right] v = -\frac{55824935636320127}{3240648032400000000} < 0.$$

Hence both individual assertions in the source conjecture have been refuted.

Task 2. Determine the optimal replacement constants

Find the smallest dimension-independent constants in (1) and (2) for each prescribed set of spectral bounds, and compare them with the weaker constants already established in the source.

Task 3. Identify maps for which the proposed bounds survive

Characterize structural conditions on Φ and on the inputs that permit the two unsquared inequalities to be squared with their original constants. Include commutativity of the relevant images and multiplicativity of the map.

Task 4. Determine the smallest counterexample dimensions

Decide whether the first assertion can fail for maps from $\mathbb{M}_2$ to itself, and classify the minimal input and output dimensions required for each failure.

Task 5. Study the corresponding weighted and power inequalities

Determine the valid exponents and sharp constants when the geometric mean is weighted or the outer square is replaced by a positive power.

References

[1] M. S. Moslehian and X. Fu, *Squaring operator Pólya–Szegő and Diaz–Metcalf type inequalities*, Linear Algebra Appl. 491 (2016), 73–82. arXiv:1501.02939v1, Conjecture 2.16, p. 9.